Problem Solving using 3D Vectors

Exam code: 9MA0

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Problem solving using 3D vectors

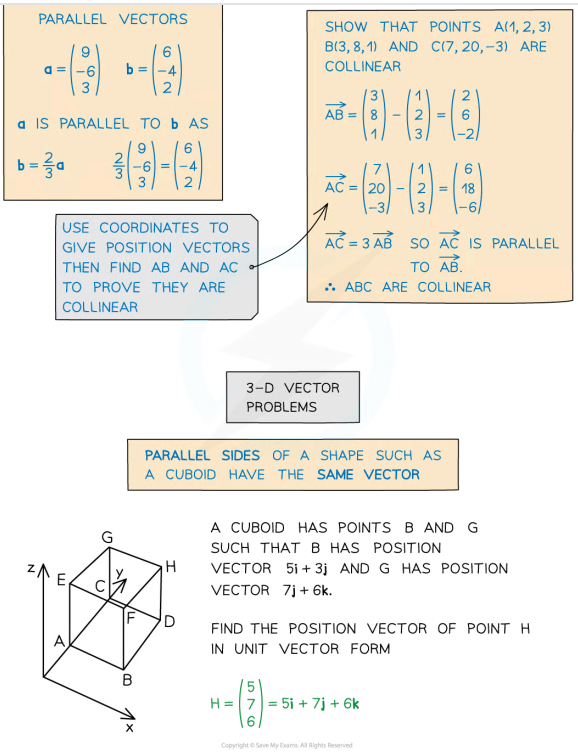
How are 3D vectors used for problem solving?

- 3D vector problems can be solved using the same principles as **2D vector problems** (see Problem Solving using Vectors)
- Vectors can be used to prove two lines are **parallel**, to show points are **collinear** (lie on the same straight line) or to find missing vertices of a given shape

Stuck on something?

Ask questions and get help with exactly what you're revising.

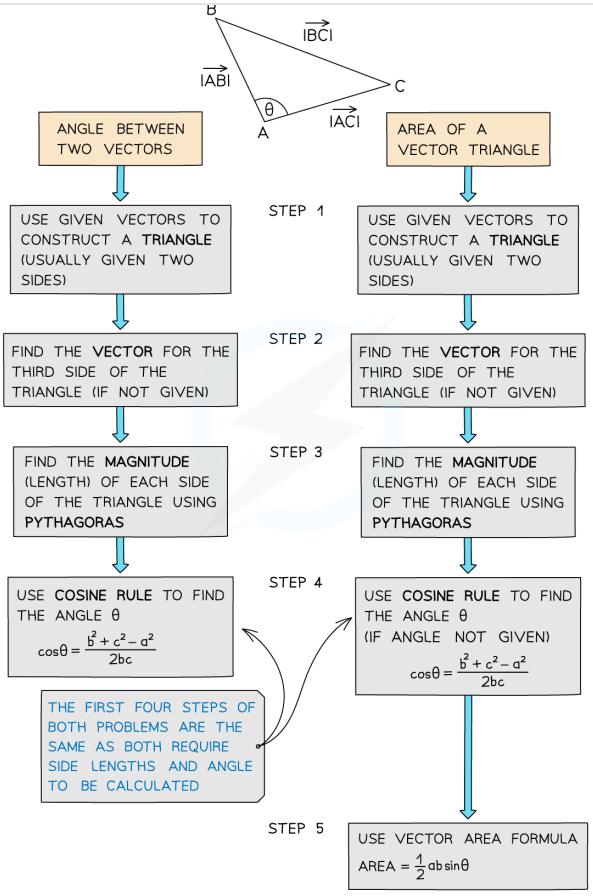
Got it



When do I use trigonometry in 3D vector problems?

- 3D vector problems can also involve using trigonometry to:
  - find the **angle between two vectors** using **Cosine Rule**
  - find the **area of a triangle** using a variation of **Area Formula**

Help me understand



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- You can find the angle between a vector and any one of the coordinate axes by using the following formulae

IF θ_x, θ_y AND θ_z ARE RESPECTIVELY THE ANGLES THAT THE VECTOR $\mathbf{a} = x\mathbf{i} + y\mathbf{j} + z\mathbf{k}$ MAKES WITH THE x–, y– AND z–AXES, THEN

$$\cos\theta_x = \frac{x}{|\mathbf{a}|}$$

$$\cos\theta_y = \frac{y}{|\mathbf{a}|}$$

$$\cos\theta_z = \frac{z}{|\mathbf{a}|}$$

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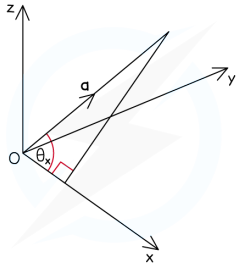

IN THE DIAGRAM ABOVE FOR EXAMPLE, THE LENGTH OF THE SIDE ADJACENT TO ANGLE θ_x IS SIMPLY THE **x** COMPONENT OF THE VECTOR $\mathbf{a} = x\mathbf{i} + y\mathbf{j} + z\mathbf{k}$, WHILE THE LENGTH OF THE HYPOTENUSE IS $|\mathbf{a}|$.

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- These formulae are not in the exam formulae book
- However, they may be derived using SOHCAHTOA as shown in the diagram
- The formulae also work for vectors in two dimensions

Examiner Tips and Tricks

- If there is a diagram, labelling all known vectors and quantities will help, and don't worry about trying to make your diagram 3-D, as long as you label it well.

Worked Example

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